Jules Streit

Robotics MSc student at EPFL with experience across the full robotic stack — wiring, debugging, and assembling hardware, embedded firmware (Arduino), computer vision, and AI-based control. Combines classical control theory (MPC, PID) with modern ML approaches, from drone flight controllers to learning-based manipulation.

jules.simonstreit@epfl.ch +33782710597 Lausanne, Switzerland

linkedin.com/in/jules-streit-55a111228 French

EDUCATION

Master of Robotics — 09/2024 – Present
École Polytechnique Fédérale de Lausanne — Lausanne
Major in robotics and minor in Space technologies, second year.

Microengineering bachelor — 09/2021 – 06/2024
École Polytechnique Fédérale de Lausanne — Lausanne
First two years at EPFL and third year at KTH (Stockholm) as an exchange student

ROBOTICS PROJECTS

Drone projects, RL & Sim2Real transfer — 02/2025 – 05/2026
Laboratory of Intelligent Systems, Pr. Dario Floreano.
- Conducted a RL study on policy architectures for an avian-inspired drone, evaluating different training strategies to improve sim-to-real transfer and policy robustness.
- Implemented gate detection and path-planning algorithms for an autonomous quadcopter drone using simulation and real-world testing. PID tuning.

Imitation Learning and navigation (Ongoing Project) — 02/2026 – Present
LeKiwi robot, EPFL AI team
- Assembled and wired electromechanical components following the LeRobot/LeKiwi open-source hardware design, including 3D-printed parts and motor integration.
- Developing a ROS2 architecture for autonomous navigation and robotic manipulation, integrating RGB cameras and LiDAR sensors with SLAM, including full hardware integration and testing with Docker-based deployment.
- Learning-based control: Developing imitation learning pipelines for robotic arm control and perception-guided manipulation.

Control of the tracking of a X-band antenna — 09/2025 – 01/2026
EPFL Spacecraft team, ESA Fly Your Satellite!
- Motor control: Programmed Arduino-based control system for dual-motor precision tracking, handling hardware integration and testing of electronics components and debugging real-time control issues.

Model Predictive Control 09/2024 – 12/2024

- Investigated advanced MPC strategies (linear, robust, and nonlinear) for car automotive cruise and lane-change control. Designed and validated controllers through simulations in MATLAB/CasADi.

Computer Vision: Image and Pattern Recognition 04/2026 – 05/2026

- Implemented end-to-end computer vision pipelines for segmentation, feature extraction, and object recognition, with emphasis on model evaluation and robustness analysis.

Deep Learning for Autonomous Vehicle 03/2026 – 05/2026

- Developed deep learning models for perception, prediction, and path planning.

R&D Mechanical Design and Prototyping 09/2024 – 06/2025

EPFL Watchmakers

- Design and prototyping of a watch mechanism on a clock scale (CAD on Autodesk Fusion 360 + 3D printing).

LANGUAGES

French ●●●●●
Mother tongue

English ●●●●○
TOEIC C1

SKILLS

Programming
Python and C++

AI-based control
Reinforcement Learning, Imitation Learning, Sim-to-Real Transfer

Deep Learning
CNN, Pytorch

Robotics
Control theory, ROS2, Direct and Inverse Kinematics, Dynamics, Motion Control, Coordinate Transformations

Hardware, Firmware & Testing
3D Printing, Motor control, Arduino, Electronics, Mechanical Assembly, Hardware-in-the-loop, Debugging

Model Predictive Control (MPC)

Tools
MATLAB/Simulink, Git, Linux, Jupyter, Docker

Computer Vision
Feature extraction, Segmentation, Object detection, OpenCV

CERTIFICATIONS

TOEIC, C1 level
Test Of English for International Communication. Listening, Reading, and Writing Comprehension, awarded in November 2024.

VOLUNTEER EXPERIENCE

Erasmus Student Network (ESN) 09/2024 – Present

Committee member — Lausanne

Organization of various events for Erasmus students at EPFL and UNIL (excursions, sports activities, social events, and visits).